

TALAL NAFEES

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PROFESSIONAL SUMMARY

Aspiring AI Engineer and Systems Developer currently entering the final year of a Bachelor's in Artificial Intelligence at the University of Sialkot. Strong practical foundation in low-level systems programming (C, Linux Kernel APIs), parallel computing (Dask), and deep learning pipeline deployment (TensorFlow, MobileNetV2, LSTM). Proven ability to combine rigorous academic engineering with operational execution.

TECHNICAL SKILLS

- **Languages:** C, Python.
- **AI/ML Frameworks:** TensorFlow/Keras, Dask Distributed, Convolutional & Recurrent Neural Networks (CNN/RNN/LSTM).
- **Systems & Tools:** Linux Kernel API, /proc virtual filesystem, Multi-threading, OS Scheduling (sched_setaffinity), Git, GitHub.
- **Concepts:** Transfer Learning, Directed Acyclic Graphs (DAG), Knowledge Representation & Reasoning (KRR).

TECHNICAL PROJECTS

Intelligent Workload Optimization System (Linux)

- **Tech Stack:** C, Linux Kernel API, Virtual Filesystem (/proc, dirent.h)
- Developed a high-performance userspace daemon in C to address resource contention and micro-stuttering in Ubuntu's Completely Fair Scheduler (CFS).
- Implemented manual kernel introspection by iterating through /proc to parse process metadata with zero overhead.
- Enforced hardware-level CPU Affinity (sched_setaffinity) to quarantine background tasks to the final core and applied dynamic priority adjustments (setpriority) to protect critical development workloads.

Smart-City CCTV Violence Detection Pipeline

- **Tech Stack:** Python, TensorFlow/Keras, MobileNetV2
- Architected an end-to-end video classification pipeline to evaluate CCTV footage across three classes (Normal, Violence, Weaponized).
- Utilized a frozen MobileNetV2 backbone (ImageNet weights) for spatial feature extraction and trained recurrent sequence models (SimpleRNN vs. LSTM).
- Achieved **98.3% test accuracy** and a 0.983 weighted F1-score using the LSTM model paired with a temporal decision rule.

Parallel Weather Data Processing Using Dask Distributed

- **Tech Stack:** Python, Dask Distributed, Pandas, DAG
- Designed a decoupled master-worker parallel processing pipeline to evaluate a synthetic dataset of 3,000,000 records.
- Benchmarked out-of-core parallel execution against sequential Pandas, achieving a **2.70× speedup factor** and a **63% runtime reduction**.
- Modeled computations as Directed Acyclic Graphs (DAG) to optimize execution paths and

manage host memory safely.

Knowledge Representation & Reasoning for Optimal Transport Selection

- **Tech Stack:** Propositional Logic, First-Order Logic (FOL), Description Logic, Resolution Proofs
- Formalized real-world decision-making scenarios across five knowledge representation paradigms.
- Validated logical consistency through Forward/Backward chaining and Resolution Proofs (CNF) to derive optimal choices under dynamic conditions.

EDUCATION

Bachelor of Science in Artificial Intelligence (BS-AI) | University of Sialkot

- Completed 6th Semester (Entering 7th Semester / Final Year)
- Roll Number: 19